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Patent Public Search | Text View

United States Patent Application Publication

20250279832

Kind Code

A1

Publication Date

September 04, 2025

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AUTOMATIC VARIABLE TILT APPARATUS, OPTICAL SIGNAL GAIN EQUALIZATION METHOD, AND NON-TRANSITORY COMPUTER READABLE MEDIUM

Abstract

An automatic variable tilt apparatus includes: a first Fi/Fo device configured to separate a multicore optical input signal into two single core optical input signals; two variable tilt equalizers configured to respectively adjust tilt levels of the two single core optical input signals; two optical couplers configured to respectively divide two single core optical output signals from the two variable tilt equalizers into two optical signals in a predetermined proportion; a feedback controller configured to perform feedback control on the associated variable tilt equalizer, based on a tilt level of one of the single core optical output signals being divided by each of the two optical couplers; and a second Fi/Fo device configured to multiplex another of the single core optical output signals being divided by each of the two optical couplers, and output a multiplexed multicore optical output signal.

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Appl. No.: 19/052348
Filed: February 13, 2025

Foreign Application Priority Data

JP 2024-029693 Feb. 29, 2024

Publication Classification

Int. Cl.: H04B10/2507 (20130101); H04B10/25 (20130101)

U.S. Cl.:

CPC H04B10/2507 (20130101); H04B10/2589 (20200501);

Background/Summary

INCORPORATION BY REFERENCE

[0001] This application is based upon and claims the benefit of priority from Japanese patent application No. 2024-029693, filed on Feb. 29, 2024, the disclosure of which is incorporated herein in its entirety by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to an automatic variable tilt apparatus and an optical signal gain equalization method that are used for a multicore fiber, and a non-transitory computer readable medium, and particularly relates to an automatic variable tilt apparatus and an optical signal gain equalization method that are used in an optical submarine system of a multicore system, and a non-transitory computer readable medium.

BACKGROUND ART

[0003] An equalization apparatus that equalizes an inclination level of an optical signal used in a submarine cable system (optical submarine system), a communication system, and an equalization method have been known (for example, see International Patent Publication No. 2019/167736). The equalization apparatus disclosed in International Patent Publication No. 2019/167736 includes an equalizer group including at least two equalizers for reducing the number of optical devices while a setting number of inclination levels is maintained.

[0004] In an optical submarine system, an optical signal in a signal bandwidth of a C band is currently used in an optical fiber of a single core (single mode fiber (SMF) on which only one core is mounted). Thus, in order to increase communication capacity of the optical submarine system, the number (core number) of optical fibers to be mounted needs to be increased.

[0005] However, in an optical submarine system in which the number of optical fibers that can be mounted on an optical submarine cable is limited due to a cross section of an optical fiber and the like, there are problems that the number of optical fibers is difficult to increase and a manufacturing cost also becomes massive.

SUMMARY

[0006] The present disclosure has been made in order to solve such problems, and an example object of the present disclosure is to provide an automatic variable tilt apparatus and an optical signal gain equalization method that can automatically adjust a tilt level (inclination) of an optical profile being an optical input signal in a short time in an optical submarine system using a multicore fiber, and provide a non-transitory computer readable medium.

[0007] In a first example aspect, an automatic variable tilt apparatus according to the present disclosure includes: [0008] a first fan-in/fan-out device configured to separate a multicore optical input signal from a multicore fiber into two single core optical input signals; [0009] two variable tilt equalizers configured to respectively adjust tilt levels of the two single core optical input signals; [0010] two optical couplers configured to respectively divide two single core optical output signals from the two variable tilt equalizers into two optical signals in a predetermined proportion; [0011] a feedback controller configured to perform feedback control on the associated variable tilt equalizer, based on a tilt level of one of the single core optical output signals being divided by each of the two optical couplers; and [0012] a second fan-in/fan-out device configured to multiplex

another of the single core optical output signals being divided by each of the two optical couplers, and output a multiplexed multicore optical output signal.

[0013] In a second example aspect, an optical signal gain equalization method according to the present disclosure includes: [0014] by a first fan-in/fan-out device, separating a multicore optical input signal from a multicore fiber into two single core optical input signals; [0015] inputting each of the two single core optical input signals to two variable tilt equalizers; [0016] by each of two optical couplers, dividing a single core optical output signal from the associated variable tilt equalizer into two optical signals in a predetermined proportion; [0017] performing feedback control on the associated variable tilt equalizer, based on a tilt level of one of the single core optical output signals being divided by each of the two optical couplers; and, [0018] by a second fan-in/fan-out device, multiplexing another of the single core optical output signals being divided by each of the two optical couplers, and outputting a multiplexed multicore optical output signal.

[0019] In a third example aspect, a non-transitory computer readable medium according to the present disclosure that stores a program causes a computer to execute: [0020] extracting an optical signal having a specific wavelength from a part of each of two single core optical input signals being separated from a multicore optical input signal by a fan-in/fan-out device; [0021] converting two optical signals having the specific wavelength into two electrical signals; [0022] analyzing a tilt level of each of the two single core optical input signals, based on the two electrical signals; [0023] each selecting one tilt equalizer from a plurality of tilt equalizers in each of two variable tilt equalizers, based on a target tilt level stored in a memory and an analysis result by the analysis, and switching to the selected one tilt equalizer; and [0024] multiplexing two optical output signals being output from the two variable tilt equalizers after the switching, and outputting a multiplexed multicore optical output signal.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0025] The above and other aspects, features, and advantages of the present disclosure will become more apparent from the following description of certain example embodiments when taken in conjunction with the accompanying drawings, in which:

[0026] FIG. 1 is a block diagram illustrating a configuration example of an automatic variable tilt apparatus according to the present disclosure;

[0027] FIGS. 2A and 2B are diagrams illustrating one example of an optical input signal to the automatic variable tilt apparatus according to the present disclosure;

[0028] FIGS. 3A and 3B are block diagrams illustrating a configuration example of variable tilt equalizers of the automatic variable tilt apparatus according to the present disclosure;

[0029] FIG. 4 is a block diagram illustrating a configuration example of a feedback controller of the automatic variable tilt apparatus according to the present disclosure;

[0030] FIG. 5 is a schematic diagram illustrating a state of an optical signal after gain equalization by the automatic variable tilt apparatus according to the present disclosure;

[0031] FIGS. 6A and 6B are block diagrams illustrating a connection example of the variable tilt equalizers after gain equalization by the automatic variable tilt apparatus according to the present disclosure;

[0032] FIG. 7 is a diagram simplifying a state of an optical signal after gain equalization by the automatic variable tilt apparatus according to the present disclosure and a connection of tilt equalizers; and

[0033] FIG. 8 is a flowchart illustrating one example of optical signal gain equalization processing performed by the automatic variable tilt apparatus according to the present disclosure.

EXAMPLE EMBODIMENT

[0034] Example embodiments of the present disclosure will be described below with reference to the drawings. However, the present disclosure in the claims is not limited to the example embodiments below. Further, all configurations described in the example embodiments are not necessarily essential as means for solving the problems. For clarification of the description, the description and the drawings below are appropriately omitted and simplified. In each of the drawings, the same elements will be denoted by the same reference signs, and duplicate description will be omitted as necessary.

[0035] Hereinafter, an automatic variable tilt apparatus for a multicore fiber according to the present example embodiment will be described with reference to FIGS. 1 to 8. Before a configuration and an operation of the automatic variable tilt apparatus for a multicore fiber (hereinafter abbreviated as the “automatic variable tilt apparatus”) are described, circumstances that led to the present disclosure will be described.

<Circumstances that LED to Present Disclosure>

[0036] As also described in the background art, in an optical submarine system in which the number of optical fibers that can be mounted on an optical submarine cable is limited, there are problems that the number of optical fibers is difficult to increase and a manufacturing cost also becomes massive. Thus, in recent years, there is a movement to use, in optical communication, a C-band multicore fiber in which a plurality of cores are mounted on one optical fiber, and practical use is also desired in the optical submarine system.

[0037] Then, in the optical submarine system at a long-distance, in order to secure transmission quality of the multicore fiber, it is desirable to make a tilt level (inclination) of a profile (optical profile) of an optical signal flat, i.e., to adjust a tilt level of an optical profile to substantially 0 dB during system operation from beginning of life (BOL) to end of life (EOL).

[0038] However, a loss of a tilt level of the optical profile is increased due to repair on an optical submarine cable and aged deterioration of the optical submarine cable. Thus, in order to keep a tilt level flat during operation of the optical submarine system, a variable tilt equalizer apparatus that can freely adjust a tilt level of the optical profile needs to be provided.

[0039] Further, operation of a multicore system has been proceeding as a multi-conductor system that requires many signal lines, but system monitoring and tilt adjustment take time due to an increase in system signal lines. As a result, there is a problem that an operation cost of an optical submarine cable using a multicore fiber increases. Furthermore, there is also a demand to automate system monitoring and tilt adjustment.

[0040] For such problems, the inventor of the present application has come up with ideas of separating a multicore optical input signal being input from a multicore fiber into a plurality of single core optical input signals for a single mode fiber, feeding back a part of an optical output signal in each single mode fiber, and switching a plurality of tilt equalizers in a variable tilt equalizer. Hereinafter, an automatic variable tilt apparatus according to the present disclosure will be described.

<Configuration of Automatic Variable Tilt Apparatus>

[0041] First, a configuration of the automatic variable tilt apparatus according to the present example embodiment will be described. The automatic variable tilt apparatus according to the present disclosure is one inserted between a multicore fiber and a multicore fiber in a multicore system. FIG. 1 is a block diagram illustrating a configuration example of an automatic variable tilt apparatus 1 according to the present disclosure. A configuration of the automatic variable tilt apparatus 1 according to the present example embodiment will be described with reference to FIG. 1.

[0042] Note that, in the present example, a case where a multicore fiber of two cores is used as a multicore fiber will be described. However, the present disclosure can also be applied to a case where a multicore fiber of three or more cores is used instead of a case where a multicore fiber of two cores is used. In that case, a processing system described below may be provided in the same

number as a core number in association with the core number.

[0043] As illustrated in FIG. 1, the automatic variable tilt apparatus **1** includes a fan-in/fan-out device (hereinafter referred to as a Fi/Fo device) **11** on an input side, two variable tilt equalizers **12** and **13**, a feedback controller **14**, two optical couplers **15** and **16**, and an Fi/Fo device **17** on an output side. A multicore optical input signal is input from a multicore fiber to the automatic variable tilt apparatus **1**. Further, a system including the variable tilt equalizer **12** is assumed to be a system A, and a system including the variable tilt equalizer **13** is assumed to be a system B.

[0044] In the present example, the multicore fiber is formed of two single core optical input signals A_i and B_i . FIGS. 2A and 2B are diagrams illustrating one example of an optical input signal to the automatic variable tilt apparatus **1** according to the present disclosure. As illustrated in FIGS. 2A and 2B, each of the single core optical input signals A_i and B_i includes a spectrum formed of five wavelengths λ_1 to λ_5 . Each of the wavelengths λ_1 to λ_5 is a center wavelength having a wavelength width in a predetermined range. Herein, the single core optical input signal A_i is assumed to have a characteristic in which a gain increases with a longer wavelength, and the single core optical input signal B_i is assumed to have a characteristic in which a gain decreases with a longer wavelength.

[0045] The Fi/Fo device **11** on the input side is configured to separate a multicore optical input signal A_i , B_i being input from the multicore fiber into the single core optical input signal A_i and the single core optical input signal B_i . Note that a signal line between the Fi/Fo device **11** and the Fi/Fo device **17** is a single mode fiber (SMF).

[0046] The single core optical input signal A_i is input to the variable tilt equalizer **12**, and the single core optical input signal B_i is input to the variable tilt equalizer **13**. The variable tilt equalizers **12** and **13** are configured to respectively adjust tilt (inclination) levels of the single core optical input signals A_i and B_i . FIGS. 3A and 3B are block diagrams illustrating a configuration example of the variable tilt equalizers **12** and **13** of the automatic variable tilt apparatus **1** according to the present disclosure.

[0047] As illustrated in FIG. 3A, the variable tilt equalizer **12** includes an optical switch (optical matrix switch) **121** on the input side, nine tilt equalizers **1201** to **1209** associated with nine kinds of gains, and an optical switch **122** on the output side. Switching among the tilt equalizers **1201** to **1209** by the optical switches **121** and **122** is controlled by the feedback controller **14**.

[0048] The nine tilt equalizers **1201** to **1209** of the variable tilt equalizer **12** include the tilt equalizer **1205** having a gain of 0 dB, and the eight tilt equalizers **1201** to **1204** and **1206** to **1209** having a gain being set at a predetermined interval for positive and negative. Herein, the predetermined interval is 1 dB. In other words, the eight tilt equalizers **1201** to **1204** and **1206** to **1209** are the tilt equalizer **1201** having a gain of -4 dB, the tilt equalizer **1202** having a gain of -3 dB, the tilt equalizer **1203** having a gain of -2 dB, the tilt equalizer **1204** having a gain of -1 dB, the tilt equalizer **1206** having a gain of +1 dB, the tilt equalizer **1207** having a gain of +2 dB, the tilt equalizer **1208** having a gain of +3 dB, and the tilt equalizer **1209** having a gain of +4 dB.

[0049] Similarly, as illustrated in FIG. 3B, the variable tilt equalizer **13** includes an optical switch **131** on the input side, nine tilt equalizers **1301** to **1309** associated with nine kinds of gains, and an optical switch **132** on the output side. Switching among the tilt equalizers **1301** to **1309** by the optical switches **131** and **132** is controlled by the feedback controller **14**.

[0050] The nine tilt equalizers **1301** to **1309** of the variable tilt equalizer **13** include the tilt equalizer **1305** having a gain of 0 dB, and the eight tilt equalizers **1301** to **1304** and **1306** to **1309** having a gain being set at a predetermined interval for positive and negative. Then, similarly to the variable tilt equalizer **12**, the eight tilt equalizers **1301** to **1304** and **1306** to **1309** are the tilt equalizer **1301** having a gain of -4 dB, the tilt equalizer **1302** having a gain of -3 dB, the tilt equalizer **1303** having a gain of -2 dB, the tilt equalizer **1304** having a gain of -1 dB, the tilt equalizer **1306** having a gain of +1 dB, the tilt equalizer **1307** having a gain of +2 dB, the tilt equalizer **1308** having a gain of +3 dB, and the tilt equalizer **1309** having a gain of +4 dB.

[0051] Note that the number of the tilt equalizers included in each of the variable tilt equalizers **12** and **13** is not limited to nine. When the number of the tilt equalizers included in each of the variable tilt equalizers **12** and **13** is generalized, the number is $2N+1$ (N is a positive integer). However, for example, N may be any of two to four in such a way as to prevent an increase in size of the entire automatic variable tilt apparatus **1**. In other words, the number of the tilt equalizers included in each of the variable tilt equalizers **12** and **13** may be any of five, seven, and nine. Further, a predetermined interval between the tilt equalizers may be appropriately adjusted in accordance with the number of the tilt equalizers.

[0052] Here, as illustrated in FIGS. **3A** and **3B**, in an initial state, the variable tilt equalizer **12** selects the tilt equalizer **1205** having the gain of 0 dB, and the variable tilt equalizer **13** selects the tilt equalizer **1305** having the gain of 0 dB. In this way, as illustrated in FIG. **1**, each of the variable tilt equalizers **12** and **13** outputs the input single core optical input signals A_i and B_i as they are.

[0053] The two optical couplers **15** and **16** are configured to respectively divide two single core optical output signals A_i and B_i from the two variable tilt equalizers **12** and **13** into two optical signals in a predetermined proportion. Here, the predetermined proportion may be, for example, 1:9 in consideration of intensity of an optical output signal on a subsequent stage, which is not limited thereto. Each of the optical couplers **15** and **16** outputs, to the feedback controller **14**, a divided optical signal in a smaller proportion being one of the divided single core optical output signals A_i and B_i . In other words, each of the optical couplers **15** and **16** outputs optical feedback signals A_f and B_f to the feedback controller **14**.

[0054] Meanwhile, each of the optical couplers **15** and **16** outputs, to the Fi/Fo device **17**, a divided optical signal in a greater proportion being another of the divided single core optical output signals A_i and B_i . In other words, each of the optical couplers **15** and **16** outputs single core optical output signals A_o and B_o to the Fi/Fo device **17**.

[0055] The Fi/Fo device **17** on the output side is configured to multiplex the single core optical output signals A_o and B_o being divided by each of the two optical couplers **15** and **16**, and output a multiplexed multicore optical output signals A_o , B_o .

[0056] The feedback controller **14** is configured to perform feedback control on the associated variable tilt equalizers **12** and **13**, based on a tilt level of the optical feedback signals A_f and B_f being divided by each of the two optical couplers **15** and **16**. Hereinafter, a configuration of the feedback controller **14** will be described with reference to FIG. **4**. FIG. **4** is a block diagram illustrating a configuration example of the feedback controller **14** of the automatic variable tilt apparatus **1** according to the present disclosure.

[0057] As illustrated in FIG. **4**, the feedback controller **14** includes a band-pass filter (BPF) **141**, a photodiode (PD) **142**, an analysis unit **143**, and a control unit **144** in the A system, a band-pass filter (BPF) **145**, a photodiode (PD) **146**, an analysis unit **147**, and a control unit **148** in the B system, and a memory **149**.

[0058] The band-pass filters **141** and **145** are configured to each extract an optical signal having a specific wavelength (herein, an optical signal having the wavelength λ_1) from the optical feedback signals A_f and B_f , and output the optical signal to the photodiodes **142** and **146**. Specifically, the band-pass filters **141** and **145** are configured to extract, as an optical signal having a specific wavelength, an optical signal having the shortest wavelength λ_1 from optical signals having the plurality of wavelengths λ_1 to λ_5 , and output the optical signal to the photodiodes **142** and **146**. The photodiodes **142** and **146** are configured to receive the optical signal having the specific wavelength λ_1 , and each convert the optical signal into two electrical signals.

[0059] The analysis units **143** and **147** are configured to respectively analyze tilt levels of the single core optical input signals A_i and B_i , based on the two electrical signals associated with the optical signal having the specific wavelength λ_1 . The memory **149** is configured to store a target tilt level. Here, the target tilt level means a tilt level at which a gain of an optical fiber is approximately fixed (0 dB) at each of the wavelengths λ_1 to λ_5 in an optical submarine system. The target tilt level is

preset in accordance with a kind of an optical fiber to be used, a transmission distance, and the like, and is stored in the memory **149**.

[0060] The control units **144** and **148** are configured to respectively select one of the plurality of tilt equalizers **1201** to **1209** and **1301** to **1309** of the associated variable tilt equalizers **12** and **13**, based on an analysis result of each of the analysis units **143** and **147** and the target tilt level stored in the memory **149**. Then, the control units **144** and **148** are configured to control the optical switches **121**, **122**, **131**, and **132** of the variable tilt equalizers **12** and **13**, based on the selected one tilt equalizer, and perform switching of the tilt equalizer.

<Operation of Terminal Apparatus>

[0061] Next, one example of an operation of the automatic variable tilt apparatus **1** according to the present example embodiment will be described with reference to FIGS. **1** to **4** described above and FIGS. **5** to **7**. In the present example embodiment, the automatic variable tilt apparatus **1** is assumed to perform an optical signal gain equalization method (optical signal gain equalization processing). Herein, an operation of each system of the A system and the B system will be described. FIG. **5** is a schematic diagram illustrating a state of an optical signal after gain equalization by the automatic variable tilt apparatus **1** according to the present disclosure. FIGS. **6A** and **6B** are block diagrams illustrating a connection example of the variable tilt equalizers after gain equalization by the automatic variable tilt apparatus **1** according to the present disclosure. FIG. **7** is a diagram simplifying a state of an optical signal after gain equalization by the automatic variable tilt apparatus **1** according to the present disclosure and a connection of the tilt equalizers.

[0062] The multicore optical input signal A_i , B_i is first input to the F_i/F_o device **11** from the multicore fiber in the multicore system, is separated into the single core optical input signal A_i and the single core optical input signal B_i by the F_i/F_o device **11**, and is divided into two single mode fibers (SMFs). First, the description of the operation of the A system continues.

[0063] In the A system, the single core optical input signal A_i is input to the optical switch **121** of the variable tilt equalizer **12**. Herein, as an initial stage, the tilt equalizer **1205** having the gain of 0 dB of the variable tilt equalizer **12** is selected, and communicates with the optical switch **122** (see FIG. **3A**). Next, the single core optical input signal A_i is output as the single core optical output signal A_i from the variable tilt equalizer **12** via the tilt equalizer **1205** and the optical switch **122** (see FIG. **1**).

[0064] Next, the single core optical output signal A_i is input to the optical coupler **15**. In the optical coupler **15**, the single core optical output signal A_i is divided into two optical signals in a predetermined proportion. A part of the divided single core optical output signal A_i is input as the optical feedback signal A_f to the feedback controller **14**.

[0065] Here, an optical signal having a specific wavelength needs to be extracted in order to acquire tilt information about the single core optical input signal A_i from the divided optical feedback signal A_f . Thus, in the feedback controller **14**, the optical feedback signal A_f is input to the band-pass filter **141**. The band-pass filter **141** extracts and outputs the optical signal having the specific wavelength λ_1 . Next, the extracted optical signal having the specific wavelength λ_1 is input to the photodiode **142**. In the photodiode **142**, the extracted optical signal having the specific wavelength λ_1 is converted into an electrical signal, and the electrical signal is output.

[0066] The electrical signal is input to the analysis unit **143**. The analysis unit **143** analyzes a tilt level of the single core optical input signal A_i , based on the electrical signal, and outputs an analysis result (analysis information). The analysis result of the analysis unit **143** is input to the control unit **144**. When the control unit **144** accepts the tilt level of the single core optical input signal A_i being the analysis result of the analysis unit **143**, the control unit **144** acquires target tilt information stored in the memory **149**.

[0067] The control unit **144** compares the tilt level of the single core optical input signal A_i with the target tilt information, and calculates a needed tilt level adjustment amount in accordance with a comparison result. In the present example, the tilt level (inclination) of the single core optical input

signal A_i is greater than a tilt level (inclination) of the single core optical input signal B_i , and the control unit **144** selects the tilt equalizer **1201** having the gain of -4 dB among the plurality of tilt equalizers **1201** to **1209** of the variable tilt equalizer **12** (see FIG. 6A). Next, the optical switches **121** and **122** of the variable tilt equalizer **12** switch from the tilt equalizer **1205** having the gain of 0 dB to the tilt equalizer **1201** having the gain of -4 dB in accordance with control of the control unit **144**, and execution of the optical signal gain equalization method ends.

[0068] When such an operation of the A system ends, and the tilt equalizer **1201** according to the tilt level of the single core optical input signal A_i is selected, a spectrum of an optical signal as indicated in the A system in FIG. 5 is acquired. In other words, a tilt of a spectrum of the single core optical input signal A_i is adjusted by the variable tilt equalizer **12**, and a single core optical output signal A_e after the tilt adjustment is output from the variable tilt equalizer **12**. Here, as illustrated in FIG. 5, the single core optical output signal A_e includes a spectrum having substantially the same gain of each of the wavelengths λ_1 to λ_5 .

[0069] Since an optical feedback signal A_{ef} divided by the optical coupler **15** also has a flat tilt level having substantially the same gain of each of the wavelengths λ_1 to λ_5 , an adjustment to the tilt level is not performed in the feedback controller **14**, and the tilt equalizer **1201** having the gain of -4 dB is still selected in the variable tilt equalizer **12**. Meanwhile, the rest of the single core optical output signal A_e after the tilt adjustment being divided by the optical coupler **15** is input, to the F_i/F_o device **17**, as a single core optical output signal A_{eo} being a part of the multicore optical output signal of the automatic variable tilt apparatus **1**.

[0070] Next, the operation of the B system will be described. In the B system, the single core optical input signal B_i is input to the optical switch **131** of the variable tilt equalizer **13**. Herein, similarly to the A system, as an initial stage, the tilt equalizer **1305** having the gain of 0 dB of the variable tilt equalizer **13** is selected, and communicates with the optical switch **132** (see FIG. 3B). Next, the single core optical input signal B_i is output as the single core optical output signal B_i from the variable tilt equalizer **13** via the tilt equalizer **1305** and the optical switch **132** (see FIG. 1).

[0071] Next, the single core optical output signal B_i is input to the optical coupler **16**. In the optical coupler **16**, the single core optical output signal B_i is divided into two optical signals in a predetermined proportion. A part of the divided single core optical output signal B_i is input as the optical feedback signal B_f to the feedback controller **14**.

[0072] Here, an optical signal having a specific wavelength needs to be extracted in order to acquire tilt information about the single core optical input signal B_i from the divided optical feedback signal B_f . Thus, in the feedback controller **14**, the optical feedback signal B_f is input to the band-pass filter **145**. The band-pass filter **145** extracts and outputs the optical signal having the specific wavelength λ_1 . Next, the extracted optical signal having the specific wavelength λ_1 is input to the photodiode **146**. In the photodiode **146**, the extracted optical signal having the specific wavelength λ_1 is converted into an electrical signal, and the electrical signal is output.

[0073] The electrical signal is input to the analysis unit **147**. The analysis unit **147** analyzes a tilt level of the single core optical input signal B_i , based on the electrical signal, and outputs an analysis result (analysis information). The analysis result of the analysis unit **147** is input to the control unit **148**. When the control unit **148** accepts the tilt level of the single core optical input signal B_i being the analysis result of the analysis unit **147**, the control unit **148** acquires target tilt information stored in the memory **149**.

[0074] The control unit **148** compares the tilt level of the single core optical input signal B_i with the target tilt information, and calculates a needed tilt level adjustment amount in accordance with a comparison result. In the present example, the tilt level (inclination) of the single core optical input signal B_i is smaller than the tilt level (inclination) of the single core optical input signal A_i , and the control unit **148** selects the tilt equalizer **1307** having the gain of $+2$ dB among the plurality of tilt equalizers **1301** to **1309** of the variable tilt equalizer **13** (see FIG. 6B). Next, the optical switches

131 and **132** of the variable tilt equalizer **13** switch from the tilt equalizer **1305** having the gain of 0 dB to the tilt equalizer **1307** having the gain of +2 dB in accordance with control of the control unit **148**, and execution of the optical signal gain equalization method ends.

[0075] When such an operation of the B system ends, and the tilt equalizer **1307** according to the tilt level of the single core optical input signal B_i is selected, a spectrum of an optical signal as indicated in the B system in FIG. 5 is acquired. In other words, a tilt of a spectrum of the single core optical input signal B_i is adjusted by the variable tilt equalizer **13**, and a single core optical output signal B_e after the tilt adjustment is output from the variable tilt equalizer **13**. Here, as illustrated in FIG. 5, the single core optical output signal B_e includes a spectrum having substantially the same gain of each of the wavelengths λ_1 to λ_5 .

[0076] Since an optical feedback signal B_{ef} divided by the optical coupler **16** also has a flat tilt level having substantially the same gain of each of the wavelengths λ_1 to λ_5 , an adjustment to the tilt level is not performed in the feedback controller **14**, and the tilt equalizer **1307** having the gain of +2 dB is still selected in the variable tilt equalizer **13**. Meanwhile, the rest of the single core optical output signal B_e after the tilt adjustment being divided by the optical coupler **16** is input, to the Fi/Fo device **17**, as a single core optical output signal B_{eo} being a part of the multicore optical output signal of the automatic variable tilt apparatus **1**.

[0077] In the end, the single core optical output signal A_{eo} being input from the optical coupler **15** and the single core optical output signal B_{eo} being input from the optical coupler **16** are multiplexed by the Fi/Fo device **17**, and a multiplexed multicore optical output signal A_{eo}, B_{eo} is output from the Fi/Fo device **17** (see FIG. 5).

[0078] For a series of the operations of the optical signal gain equalization method described above, a state of an optical signal after gain equalization by the automatic variable tilt apparatus **1** and a connection of the tilt equalizers will be briefly described by using FIG. 7. First, the multicore optical input signal A_i, B_i is input to the Fi/Fo device **11** from the multicore fiber in the multicore system, and is separated into the single core optical input signal A_i and the single core optical input signal B_i .

[0079] Next, the single core optical input signal A_i is input to the tilt equalizer **1201** having the gain of -4 dB, and the single core optical output signal A_e after the tilt adjustment having a flat tilt level is output. Meanwhile, the single core optical input signal B_i is input to the tilt equalizer **1307** having the gain of +2 dB, and the single core optical output signal B_e after the tilt adjustment having a flat tilt level is output.

[0080] In the end, although not illustrated, the single core optical output signals A_{eo} and B_{eo} being a difference between the optical feedback signals A_{ef} and B_{ef} are input to the Fi/Fo device **17**, and the multicore optical output signal A_{eo}, B_{eo} is output from the Fi/Fo device **17** after multiplexing by the Fi/Fo device **17**.

[0081] Next, an operation of the automatic variable tilt apparatus **1** will be described again with reference to a flowchart illustrated in FIG. 8. FIG. 8 is the flowchart illustrating one example of the optical signal gain equalization processing performed by the automatic variable tilt apparatus **1** according to the present disclosure.

[0082] The automatic variable tilt apparatus **1** according to the present example embodiment performs the optical signal gain equalization processing, for example, at any timing or at all times. When the optical signal gain equalization processing starts, the multicore optical input signal A_i, B_i from the multicore fiber in the multicore system is first separated by the Fi/Fo device **11** into the two single core optical input signals A_i and B_i (step S1).

[0083] Next, each of the single core optical output signals A_i and B_i being output from each of the variable tilt equalizers **12** and **13** are divided by the optical couplers **15** and **16** into two optical signals (i.e., the single core optical output signals A_o and B_o and the optical feedback signals A_f and B_f) (step S2).

[0084] Next, the optical feedback signals A_f and B_f are input to the feedback controller **14** (step

S3), and a tilt level of an optical signal having the specific wavelength $\lambda 1$ being extracted by each of the band-pass filters **141** and **145** is analyzed by the analysis units **143** and **147** in the feedback controller **14** (step S4).

[0085] Next, the tilt level being analyzed by the analysis units **143** and **147** is compared with a target tilt level being stored in the memory **149** by the control units **144** and **148** (step S5), and a needed tilt level adjustment amount is calculated in accordance with a comparison result. Then, one tilt equalizer in each of the variable tilt equalizers **12** and **13** is selected by the control units **144** and **148**, based on a comparison result (tilt level adjustment amount) (step S6).

[0086] Next, switching of a connection to the one tilt equalizer selected in each of the variable tilt equalizers **12** and **13** is performed in accordance with control of the control units **144** and **148** (step S7), and the optical signal gain equalization processing ends.

[0087] As described above, the automatic variable tilt apparatus **1** according to the present example embodiment includes: the first fan-in/fan-out device **11** configured to separate the multicore optical input signal A_i , B_i from a multicore fiber into the two single core optical input signals A_i and B_i ; the two variable tilt equalizers **12** and **13** configured to respectively adjust tilt levels of the two single core optical input signals A_i and B_i ; the two optical couplers **15** and **16** configured to respectively divide the two single core optical output signals A_i and B_i or A_e and B_e from the two variable tilt equalizers **12** and **13** into the two optical signals A_o , A_f and B_o , B_f , or A_{eo} , A_{ef} and B_{eo} , B_{ef} in a predetermined proportion; the feedback controller **14** configured to perform feedback control on the associated variable tilt equalizers **12** and **13**, based on a tilt level of one A_f , B_f of the single core optical output signals being divided by each of the two optical couplers **15** and **16**; and the second fan-in/fan-out device **17** configured to multiplex another A_o , B_o , or A_{eo} , B_{eo} of the single core optical output signals being divided by each of the two optical couplers **15** and **16**, and output the multiplexed multicore optical output signal A_o , B_o , or A_{eo} , B_{eo} . The automatic variable tilt apparatus **1** is configured in such a manner, and thus a gain of the variable tilt equalizers **12** and **13** can be changed in accordance with control of the feedback controller **14**, and a tilt level of the single core optical input signals A_i and B_i can be thus automatically adjusted to be flat. In this way, the automatic variable tilt apparatus **1** according to the present example embodiment can automatically adjust a tilt level (inclination) of an optical profile (spectrum) being an optical input signal in a short time in an optical submarine system using a multicore fiber. Then, monitoring of the optical submarine system can be simplified by automatically adjusting the optical profile (spectrum).

[0088] Further, in the automatic variable tilt apparatus **1** according to the present example embodiment, in association with each of the two single core optical output signals A_i and B_i , the feedback controller **14** includes the two band-pass filters **141** and **145** configured to each extract an optical signal having the specific wavelength $\lambda 1$ from one A_f , B_f of the single core optical output signals, and output the optical signal, the two photodiodes **142** and **146** configured to receive the two optical signals having the specific wavelength $\lambda 1$, and each convert the two optical signals into two electrical signals, the two analysis units **143** and **147** configured to respectively analyze tilt levels of the two single core optical input signals A_i and B_i , based on the two electrical signals, the memory **149** configured to store a target tilt level, and the two control units **144** and **148** configured to respectively select one tilt equalizer in each of the variable tilt equalizers **12** and **13**, based on an analysis result of the analysis units **143** and **147** and the target tilt level, and control switching of the tilt equalizer in the variable tilt equalizers **12** and **13**. Further, each of the two variable tilt equalizers **12** and **13** includes the plurality of tilt equalizers **1201** to **1209** and **1301** to **1309** associated with a plurality of gains, and the optical switches **121**, **122**, **131**, and **132** associated with the number of the plurality of tilt equalizers **1201** to **1209** and **1301** to **1309**. The automatic variable tilt apparatus **1** according to the present example embodiment is configured in such a manner, and thus a tilt level of the single core optical input signals A_i and B_i can be automatically adjusted to be flat by switching a small number of the tilt equalizers in addition to the effects described above.

[0089] Further, an optical signal gain equalization method includes: by the first fan-in/fan-out device **11**, separating the multicore optical input signal A_i , B_i from a multicore fiber into the two single core optical input signals A_i and B_i ; inputting each of the two single core optical input signals A_i and B_i to the two variable tilt equalizers **12** and **13**; by each of the two optical couplers **15** and **16**, dividing the single core optical output signals A_i and B_i from the associated variable tilt equalizers **12** and **13** into two optical signals in a predetermined proportion; performing feedback control on the associated variable tilt equalizers **12** and **13**, based on a tilt level of one A_f , B_f of the single core optical output signals being divided by each of the two optical couplers **15** and **16**; and, by the second fan-in/fan-out device **17**, multiplexing another A_{eo} , B_{eo} of the single core optical output signals being divided by each of the two optical couplers **15** and **16**, and outputting the multiplexed multicore optical output signal A_{eo} , B_{eo} . The optical signal gain equalization method is configured in such a manner, and thus an effect similar to that of the automatic variable tilt apparatus **1** according to the present example embodiment can be achieved.

[0090] A part or the whole of the processing in the feedback controller **14**, particularly the analysis units **143** and **147** and the control units **144** and **148** of the automatic variable tilt apparatus **1** described above can be achieved as a computer program that can be installed in the automatic variable tilt apparatus **1**. Such a program can be stored and provided to a computer using any type of non-transitory computer readable media. Non-transitory computer readable media include any type of tangible storage media. Examples of non-transitory computer readable media include magnetic storage media (such as floppy disks, magnetic tapes, hard disk drives, etc.), optical magnetic storage media (e.g. magneto-optical disks), CD-ROM (compact disc read only memory), CD-R (compact disc recordable), CD-R/W (compact disc rewritable), and semiconductor memories (such as mask ROM, PROM (programmable ROM), EPROM (erasable PROM), flash ROM, RAM (random access memory), etc.). The program may be provided to a computer using any type of transitory computer readable media. Examples of transitory computer readable media include electric signals, optical signals, and electromagnetic waves. Transitory computer readable media can provide the program to a computer via a wired communication line (e.g. electric wires, and optical fibers) or a wireless communication line.

[0091] The present disclosure can provide an automatic variable tilt apparatus, an optical signal gain equalization method, and a non-transitory computer readable medium that stores a program, which can automatically adjust a tilt level (inclination) of an optical profile being an optical input signal in a short time in an optical submarine system using a multicore fiber.

[0092] While the disclosure has been particularly shown and described with reference to embodiments thereof, the disclosure is not limited to these embodiments. It will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope of the present disclosure as defined by the claims.

[0093] A part or the whole of the above-mentioned example embodiments may be described in supplementary notes below, which is not limited thereto.

(Supplementary Note 1)

[0094] An automatic variable tilt apparatus including: [0095] a first fan-in/fan-out device configured to separate a multicore optical input signal from a multicore fiber into two single core optical input signals; [0096] two variable tilt equalizers configured to respectively adjust tilt levels of the two single core optical input signals; [0097] two optical couplers configured to respectively divide two single core optical output signals from the two variable tilt equalizers into two optical signals in a predetermined proportion; [0098] a feedback controller configured to perform feedback control on the associated variable tilt equalizer, based on a tilt level of one of the single core optical output signals being divided by each of the two optical couplers; and [0099] a second fan-in/fan-out device configured to multiplex another of the single core optical output signals being divided by each of the two optical couplers, and output a multiplexed multicore optical output signal.

(Supplementary Note 2)

[0100] The automatic variable tilt apparatus according to supplementary note 1, wherein [0101] each of the two variable tilt equalizers includes a plurality of tilt equalizers associated with a plurality of gains, and an optical switch associated with the number of the plurality of tilt equalizers, [0102] the feedback controller selects one tilt equalizer from the plurality of tilt equalizers of each of the two variable tilt equalizers, based on a tilt level of one of the associated single core optical output signals, and [0103] the optical switch of each of the two variable tilt equalizers switches to the selected one tilt equalizer.

(Supplementary Note 3)

[0104] The automatic variable tilt apparatus according to supplementary note 2, wherein, [0105] in association with each of the two single core optical output signals, the feedback controller includes [0106] two band-pass filters configured to each extract an optical signal having a specific wavelength from one of the single core optical output signals, and output the optical signal, [0107] two photodiodes configured to receive the two optical signals having the specific wavelength, and each convert the two optical signals into two electrical signals, [0108] two analysis units configured to respectively analyze tilt levels of the two single core optical input signals, based on the two electrical signals, [0109] a memory configured to store a target tilt level, and [0110] two control units configured to respectively select the one tilt equalizer, based on an analysis result of the analysis unit and the target tilt level, and control switching of the tilt equalizer in the variable tilt equalizer.

(Supplementary Note 4)

[0111] The automatic variable tilt apparatus according to supplementary note 3, wherein [0112] each of the two single core optical input signals is formed of optical signals having a plurality of wavelengths, and [0113] the two band-pass filters extract, as an optical signal having the specific wavelength, an optical signal having the shortest wavelength from the optical signals having the plurality of wavelengths.

(Supplementary Note 5)

[0114] The automatic variable tilt apparatus according to supplementary note 3, wherein each of the two control units compares an analysis result of the associated analysis unit with the target tilt level, calculates a tilt level adjustment amount, and selects the one tilt equalizer, based on the tilt level adjustment amount.

(Supplementary Note 6)

[0115] The automatic variable tilt apparatus according to any one of supplementary notes 2 to 5, wherein the plurality of tilt equalizers include a tilt equalizer having a gain of 0 dB, and 2N (N is a positive integer) tilt equalizers having a gain being set at a predetermined interval for positive and negative.

(Supplementary Note 7)

[0116] The automatic variable tilt apparatus according to supplementary note 6, wherein the N is any of two to four.

(Supplementary Note 8)

[0117] An optical signal gain equalization method including: [0118] by a first fan-in/fan-out device, separating a multicore optical input signal from a multicore fiber into two single core optical input signals; [0119] inputting each of the two single core optical input signals to two variable tilt equalizers; [0120] by each of two optical couplers, dividing a single core optical output signal from the associated variable tilt equalizer into two optical signals in a predetermined proportion; [0121] performing feedback control on the associated variable tilt equalizer, based on a tilt level of one of the single core optical output signals being divided by each of the two optical couplers; and, [0122] by a second fan-in/fan-out device, multiplexing another of the single core optical output signals being divided by each of the two optical couplers, and outputting a multiplexed multicore optical output signal.

(Supplementary Note 9)

[0123] The optical signal gain equalization method according to supplementary note 8, wherein [0124] each of the two variable tilt equalizers includes a plurality of tilt equalizers associated with a plurality of gains, and an optical switch associated with the number of the plurality of tilt equalizers, and, [0125] in the feedback control, the optical signal gain equalization method further includes selecting one tilt equalizer from the plurality of tilt equalizers of each of the two variable tilt equalizers, based on a tilt level of one of the associated single core optical output signals, and, by the optical switch, switching to the selected one tilt equalizer.

(Supplementary Note 10)

[0126] The optical signal gain equalization method according to supplementary note 9, further including: [0127] in the feedback control, in association with each of the two single core optical input signals, [0128] extracting an optical signal having a specific wavelength from one of the single core optical output signals; [0129] converting two optical signals having the specific wavelength into two electrical signals; [0130] analyzing a tilt level of each of the two single core optical input signals, based on the two electrical signals; and [0131] each selecting the one tilt equalizer, based on a target tilt level stored in a memory and an analysis result by the analysis, and switching to the selected one tilt equalizer.

(Supplementary Note 11)

[0132] The optical signal gain equalization method according to supplementary note 10, wherein [0133] each of the two single core optical input signals is formed of optical signals having a plurality of wavelengths, and [0134] the optical signal gain equalization method further includes extracting, as an optical signal having the specific wavelength, an optical signal having the shortest wavelength from the optical signals having the plurality of wavelengths.

(Supplementary Note 12)

[0135] The optical signal gain equalization method according to supplementary note 10, further including: [0136] in the step of switching to the selected one tilt equalizer, [0137] comparing an analysis result by the analysis with the target tilt level, and calculating a tilt level adjustment amount; and [0138] selecting the one tilt equalizer, based on the tilt level adjustment amount.

(Supplementary Note 13)

[0139] A non-transitory computer readable medium that stores a program causing a computer to execute: [0140] extracting an optical signal having a specific wavelength from a part of each of two single core optical input signals being separated from a multicore optical input signal by a fan-in/fan-out device; [0141] converting two optical signals having the specific wavelength into two electrical signals; [0142] analyzing a tilt level of each of the two single core optical input signals, based on the two electrical signals; [0143] each selecting one tilt equalizer from a plurality of tilt equalizers in each of two variable tilt equalizers, based on a target tilt level stored in a memory and an analysis result by the analysis, and switching to the selected one tilt equalizer; and [0144] multiplexing two optical output signals being output from the two variable tilt equalizers after the switching, and outputting a multiplexed multicore optical output signal.

Claims

1. An automatic variable tilt apparatus comprising: a first fan-in/fan-out device configured to separate a multicore optical input signal from a multicore fiber into two single core optical input signals; two variable tilt equalizers configured to respectively adjust tilt levels of the two single core optical input signals; two optical couplers configured to respectively divide two single core optical output signals from the two variable tilt equalizers into two optical signals in a predetermined proportion; a feedback controller configured to perform feedback control on the associated variable tilt equalizer, based on a tilt level of one of the single core optical output signals being divided by each of the two optical couplers; and a second fan-in/fan-out device configured to multiplex another of the single core optical output signals being divided by each of the two optical

couplers, and output a multiplexed multicore optical output signal.

2. The automatic variable tilt apparatus according to claim 1, wherein each of the two variable tilt equalizers includes a plurality of tilt equalizers associated with a plurality of gains, and an optical switch associated with the number of the plurality of tilt equalizers, the feedback controller selects one tilt equalizer from the plurality of tilt equalizers of each of the two variable tilt equalizers, based on a tilt level of one of the associated single core optical output signals, and the optical switch of each of the two variable tilt equalizers switches to the selected one tilt equalizer.

3. The automatic variable tilt apparatus according to claim 2, wherein, the feedback controller includes, in association with each of the two single core optical output signals, two band-pass filters configured to each extract an optical signal having a specific wavelength from one of the single core optical output signals, and output the optical signal, two photodiodes configured to receive two optical signals having the specific wavelength, and convert the two optical signals into two electrical signals, two analysis units configured to respectively analyze tilt levels of the two single core optical input signals, based on the two electrical signals, a memory configured to store a target tilt level, and two control units configured to respectively select the one tilt equalizer, based on an analysis result of the analysis unit and the target tilt level, and control switching of the tilt equalizer in the variable tilt equalizer.

4. The automatic variable tilt apparatus according to claim 3, wherein each of the two single core optical input signals is formed of optical signals having a plurality of wavelengths, and the two band-pass filters extract, as an optical signal having the specific wavelength, an optical signal having the shortest wavelength from the optical signals having the plurality of wavelengths.

5. The automatic variable tilt apparatus according to claim 3, wherein each of the two control units compares an analysis result of the associated analysis unit with the target tilt level, calculates a tilt level adjustment amount, and selects the one tilt equalizer, based on the tilt level adjustment amount.

6. The automatic variable tilt apparatus according to claim 2, wherein the plurality of tilt equalizers include a tilt equalizer having a gain of 0 dB, and $2N$ (N is a positive integer) tilt equalizers having a gain being set at a predetermined interval for positive and negative.

7. The automatic variable tilt apparatus according to claim 6, wherein the N is any of two to four.

8. An optical signal gain equalization method comprising: by a first fan-in/fan-out device, separating a multicore optical input signal from a multicore fiber into two single core optical input signals; inputting each of the two single core optical input signals to two variable tilt equalizers; by each of two optical couplers, dividing a single core optical output signal from the associated variable tilt equalizer into two optical signals in a predetermined proportion; performing feedback control on the associated variable tilt equalizer, based on a tilt level of one of the single core optical output signals being divided by each of the two optical couplers; and, by a second fan-in/fan-out device, multiplexing another of the single core optical output signals being divided by each of the two optical couplers, and outputting a multiplexed multicore optical output signal.

9. The optical signal gain equalization method according to claim 8, wherein each of the two variable tilt equalizers includes a plurality of tilt equalizers associated with a plurality of gains, and an optical switch associated with the number of the plurality of tilt equalizers, and, in the feedback control, the optical signal gain equalization method further comprises selecting one tilt equalizer from the plurality of tilt equalizers of each of the two variable tilt equalizers, based on a tilt level of one of the associated single core optical output signals, and, by the optical switch, switching to the selected one tilt equalizer.

10. The optical signal gain equalization method according to claim 9, further comprising: in the feedback control, in association with each of the two single core optical input signals, extracting an optical signal having a specific wavelength from one of the single core optical output signals; converting two optical signals having the specific wavelength into two electrical signals; analyzing a tilt level of each of the two single core optical input signals, based on the two electrical signals;

and each selecting the one tilt equalizer, based on a target tilt level stored in a memory and an analysis result by the analysis, and switching to the selected one tilt equalizer.

11. The optical signal gain equalization method according to claim 10, wherein each of the two single core optical input signals is formed of optical signals having a plurality of wavelengths, and the optical signal gain equalization method further comprises extracting, as an optical signal having the specific wavelength, an optical signal having the shortest wavelength from the optical signals having the plurality of wavelengths.

12. The optical signal gain equalization method according to claim 10, further comprising: in the step of switching to the selected one tilt equalizer, comparing an analysis result by the analysis with the target tilt level, and calculating a tilt level adjustment amount; and selecting the one tilt equalizer, based on the tilt level adjustment amount.

13. A non-transitory computer readable medium that stores a program causing a computer to execute: extracting an optical signal having a specific wavelength from a part of each of two single core optical input signals being separated from a multicore optical input signal by a fan-in/fan-out device; converting two optical signals having the specific wavelength into two electrical signals; analyzing a tilt level of each of the two single core optical input signals, based on the two electrical signals; each selecting one tilt equalizer from a plurality of tilt equalizers in each of two variable tilt equalizers, based on a target tilt level stored in a memory and an analysis result by the analysis, and switching to the selected one tilt equalizer; and multiplexing two optical output signals being output from the two variable tilt equalizers after the switching, and outputting a multiplexed multicore optical output signal.
